

Structured Sparse mmWave 3-D SAR Imaging via Truncated-DCT and Toeplitz Matrix Methods

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High-resolution millimeter-wave (mmWave) 3-D synthetic aperture radar (SAR) imaging typically requires dense spatial sampling over large arrays or structured scanning trajectories. This substantially increases hardware cost and scanning times. Sparse reconstruction approaches based on compressed sensing (CS) and matrix completion (MC) have been explored to alleviate this burden. However, existing methods mainly rely on random sparse sampling or calibrated scanning patterns that limit practical deployment. To address this, we implement a mmWave 3-D SAR imaging system and propose a structured sparse scanning trajectory, termed expandable uniform row (EUR, which effectively reduces scanning positions and accelerates data acquisition. An efficient MC-based algorithm is then developed to reconstruct the structured sparse SAR data acquired with EUR. The algorithm first incorporates a truncated discrete cosine transform (truncated-DCT) to compress the preliminary data and enhance the reconstruction efficiency. It further employs a class of Toeplitz matrix methods for data recovery, where two Toeplitz transformations are applied to the entire matrix to handle structured sparsity. The lifted

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Toeplitz matrices are subsequently solved within a factorized alternating direction method of multipliers (ADMM) framework, avoiding singular value decomposition (SVD) and large-scale matrix inversion. Extensive experiments demonstrate that the proposed algorithm achieves high-fidelity reconstruction and imaging within 5 s for a 100×201 echo matrix across various objects and sampling rates, enabling up to a 75% reduction in data acquisition time with the EUR trajectory.

I. INTRODUCTION

Millimeter-wave (mmWave) radar can propagate through optically opaque media and penetrate nonmetallic materials, while providing high resolution at low radiated power. These advantages facilitate its adoption in nondestructive testing [1], through-the-wall imaging [2], public-security screening [3], and medical diagnosis [4].

There are two primary implementations for mmWave 3-D synthetic aperture radar (SAR) imaging. One is deploying massive multiple-input-multiple-output (MIMO) mmWave radar arrays [5], [6], and the other is performing a 2-D scan with a single-input-single-output (SISO) radar to emulate MIMO operation [7]. The former suffers from high hardware complexity and power cost [8]. The latter, with reduced hardware costs, is plagued by increased temporal overhead [9]. To strike a balance between the MIMO hardware complexity and the SISO scanning time, various trade-off strategies have been explored. Yanik et al. [10] replaced a subset of the mechanical scanning with cascaded sensors to emulate a small-scale extended MIMO configuration. However, the resulting multistatic geometries are difficult to calibrate and commonly lead to imaging artifacts [11]. Alternatively, Shao et al. [12] proposed a quasi-multistatic scanning architecture that reduces data acquisition time and established a corresponding image quality evaluation framework [13]. Nevertheless, the associated hardware cost remains relatively high. To overcome these limitations, sparse recovery strategies have been extensively adopted to reduce the amount of echo data required for high-resolution imaging, with prominent approaches based on compressed sensing (CS) and matrix completion (MC).

CS is a foundational and effective method for sparse reconstruction [14]. It formulates SAR imaging as an optimization problem that leverages the inherent sparsity of the signal or its representation in a predefined dictionary. For example, Wang et al. [15] integrated CS with deep unfolding networks and achieved fast SAR imaging under sparse sampling. Zhang et al. [16] combined the iterative shrinkage-thresholding algorithm (ISTA) with the range migration algorithm (RMA) to enhance the resolution and suppress background noise of auto-SAR images. Furthermore, Chen et al. [17] and Song et al. [18] adopted the alternating direction method of multipliers (ADMM) [19] to solve the CS optimization problem and facilitate efficient SAR reconstructions. Although CS-based methods are theoretically effective, their reliance on uniformly random sparse (URS) sampling poses challenges for practical implementation in real-world SAR systems, which typically operate along structured motion trajectories. In addition, the performance of CS-based methods relying on discretized

sparse models can be degraded by mismatches between the assumed measurement matrix and the physical scene, commonly referred to as the off-grid problem.

As a robust and advanced image processing technique, MC estimates missing entries by exploiting intrinsic data correlations within a matrix that satisfies low-rank constraints [20], [21]. Building on this property, MC is naturally immune to the grid mismatch [22] and has been widely investigated for the reconstruction of low-rank SAR data from limited observations [22], [23], [24]. However, these approaches are still developed under URS sampling, and struggle with the structured missingness arising in practical SAR acquisitions. Recently, a variety of techniques have been introduced to overcome this problem. In [25], a rotational expansion is proposed to mitigate the structured missingness, followed by the singular value thresholding (SVT) [26] for sparse SAR reconstruction. However, the rotated-expansion increases dimensionality and SVT suffers from slow convergence. To improve computational efficiency, Liu et al. [27] combined the fast-converging inexact augmented Lagrangian multiplier (IALM) [28] with the Hankel transformation [29] to recover structured missing SAR data with acceptable complexity. Such Hankel MC methods are well suited for randomly clustered missing data [27] but have limitations for uniformly structured sparse reconstruction [30], [31]. Moreover, the anti-diagonal structure of the Hankel matrix increases the computational cost in iterative optimization [32], [33], [34]. As the structural dual of Hankel matrix, the Toeplitz matrix offers improved computational efficiency due to its diagonal structure and exhibits enhanced low-rank property when representing shift-invariant signals [35], [36], making it particularly suitable for radar applications [37], [38]. Leveraging these advantages, Hai et al. [39] proposed a zigzag sparse scanning strategy combined with the Toeplitz-based Truncated Schatten-p Norm and sparse regularizer (TSPN-SR) for structured sparse data recovery. While the method effectively reduces data acquisition time, it requires precise trajectory calibration and column-by-column Toeplitz transformations, which may increase computational complexity.

In summary, current sparse reconstruction approaches mainly rely on random sparse sampling or calibrated scanning patterns, which are difficult to realize in practical SAR systems constrained by structured motion trajectories. Moreover, they exhibit reduced robustness and high computational complexity when handling structured sparse SAR data. To tackle these challenges, this article presents and validates a structured sparse trajectory design in a practical mmWave 3-D SAR imaging system, along with an efficient MC-based reconstruction algorithm that jointly exploits the truncated discrete cosine transform (truncated-DCT) and Toeplitz matrix methods. The main contributions of this work are as follows.

- 1) We propose a physically realizable expandable uniform row (EUR) scanning strategy to replace theoretical sparse sampling or calibrated scanning

patterns. This strategy significantly reduces data acquisition time required for high-resolution imaging while maintaining compatibility with mechanical scanning systems.

- 2) We develop an efficient MC-based reconstruction algorithm that introduces two novel matrix-level Toeplitz constructions to address EUR-induced structured sparsity, enabling more effective exploitation of intrinsic data correlations than existing column-wise Toeplitz or rotated-expansion approaches.
- 3) To improve computational efficiency, we incorporate a truncated-DCT at the front end of the algorithm to reduce the problem dimensionality. We also solve the lifted Toeplitz matrices within a factorized ADMM framework to avoid singular value decomposition (SVD) and large-scale matrix inversion.
- 4) The proposed scanning strategy and reconstruction algorithm are validated on a practical mmWave 3-D SAR imaging system. Experiments demonstrate accurate reconstruction and high-quality imaging within 5 s for a 100×201 echo matrix across various targets and sampling rates, achieving up to a 75% reduction in data acquisition time compared with full sampling.

The rest of this article is organized as follows. Section II establishes the mmWave SAR imaging model and data acquisition scheme, Section III details the proposed reconstruction algorithm and its theoretical underpinnings. Section IV describes the experimental setup and provides comparative results. Finally, Section V concludes this article.

II. IMAGING SCHEME AND SIGNAL MODEL

A. Imaging Prototype and Scanning Strategy

We first detail the typical mmWave 3-D imaging prototype and the full sampling scheme for data acquisition. On this basis, we subsequently introduce the proposed EUR scanning strategy.

1) *Imaging Prototype and Full Sampling Pattern:* In this work, a mmWave radar is translated at a constant speed in the XOY plane to collect data over a rectangular grid of measurement points, as illustrated in Fig. 1. A Cartesian coordinate system is adopted, where the X-axis, Y-axis, and Z-axis denote the vertical, horizontal, and range directions, respectively. The resulting synthetic aperture spans an area of $D_x \times D_y$ with sampling intervals d_x and d_y . The scan plane is fixed at $Z = 0$, and the virtual transceiver is located at grid point $(x_m, y_n, 0)$. The imaged target P is modeled as a collection of scattering points, where the p -th scatterer is located at (x_p, y_p, z_p) .

The fully sampled echo is illustrated in Fig. 2, and can be interpreted as a sequence of 2-D sampled fast time instances, represented by the small blue points within the green box. The 3-D RMA [7] is then employed for fast imaging. First, range-based pulse compression is applied to

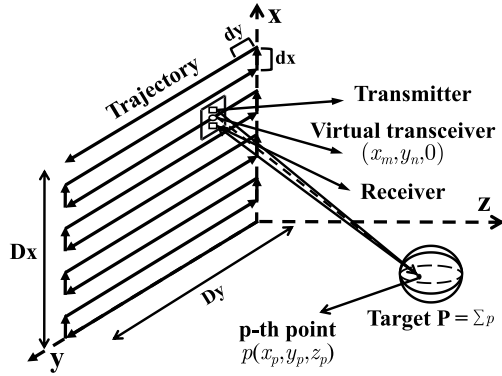


Fig. 1. 3-D mmWave SAR imaging geometry.

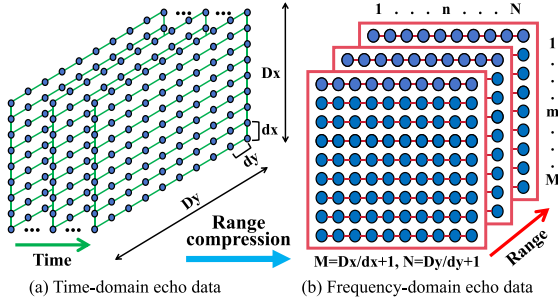


Fig. 2. Fully sampled echo data. (a) Time-domain slices. (b) Frequency-domain slices.

transform the echo data into frequency-domain range slices, each forming a $M \times N$ matrix corresponding to the enlarged blue dots within the red box. Subsequently, the 3-D image is obtained by applying 2-D RMA to each slice and coherently superimposing the results [39].

2) *Structured Sparse Scanning Strategy*: To improve imaging efficiency and reduce acquisition time in practice, we propose an EUR scanning strategy. Unlike the full sampling scheme in Fig. 2, the proposed EUR strategy adopts a two-stage row-wise undersampling and expansion mechanism. Specifically, the echo data are first acquired using uniform row-wise sub-Nyquist sampling with an enlarged vertical spacing, and then embedded into an expanded $M \times N$ echo matrix S_e via interpolation along the row dimension.

For a given sampling rate r_s , the number of sampled rows is $N_r = \lceil M \times r_s \rceil$, and their indices in S_e are given by $I = \{1, 1 + 1/r_s, \dots, 1 + (N_r - 1)/r_s\}$, while the remaining rows are set to zero and further recovered via MC. This strategy enables a controllable tradeoff between acquisition time and imaging resolution. In this article, two representative configurations with $r_s = 50\%$ and $r_s = 25\%$ are considered, as depicted in Fig. 3.

B. Signal Model and Low-Rank Property

Section II-A introduced the imaging geometry and the scanning strategy EUR. Leveraging the frequency-modulated continuous wave (FMCW) signal model of the employed mmWave radar, this subsection investigates the

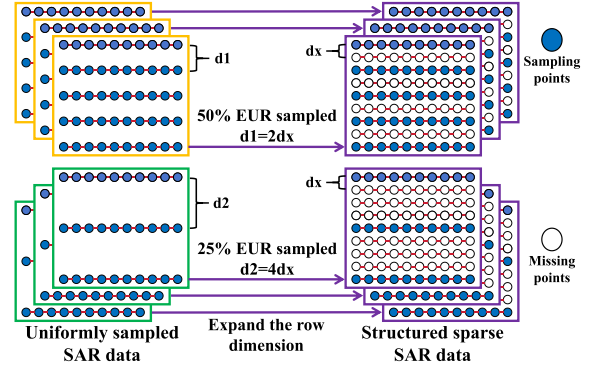


Fig. 3. EUR sampling configuration.

low-rank property of the structured sparse data acquired with EUR, as it is essential for MC.

1) *Echo Signal Model*: Consider the complex form of the FMCW signal transmitted by the radar

$$s_{tx}(t) = \exp \left\{ j2\pi \left(f_0 t + \frac{1}{2} K t^2 \right) \right\}, 0 \leq t \leq T \quad (1)$$

where f_0 is the carrier frequency at time $t = 0$, $K = B/T$ is the slope of frequency computed from the sweep bandwidth B and the signal duration T . Subsequently, the received signal is mixed with the transmitted signal to generate the intermediate-frequency (IF) signal. Incorporating the assumptions from Section II-A, we can derive the IF signal form for the radar at the scanning point $(x_m, y_n, 0)$ with respect to the p -th scattering point $p(x_p, y_p, z_p)$

$$s_{IF}(m, n, p, t) = \sigma_p \exp \left\{ j2\pi (f_0 \tau_p + K \tau_p t) \right\} \quad (2)$$

where σ_p denotes the backscattering coefficient of point p , $K \tau_p$ corresponds to the beat frequency of point p , $\tau_p = \frac{2R_p}{c}$ is the round-trip delay calculated by the distance R_p and the speed of light c . For a relatively large imaging distance, R_p can be approximated as follows [22], [39]:

$$\begin{aligned} R_p &= \sqrt{(x_m - x_p)^2 + (y_n - y_p)^2 + z_p^2} \\ &\approx R_0 + \frac{(x_m)^2 - 2x_m x_p}{2R_0} + \frac{(y_n)^2 - 2y_n y_p}{2R_0} \end{aligned} \quad (3)$$

where $R_0 = \sqrt{x_p^2 + y_p^2 + z_p^2}$ represents the distance from p to the center of the 2-D scanning plane. For a given scanning point $(x_m, y_n, 0)$, the IF signal originates from all the scattering points and can be presented as

$$s_{IF}(m, n, t) = \sum_{p=1}^{N_p} \sigma_p \exp \left\{ j2\pi (f_0 \tau_p + K \tau_p t) \right\}. \quad (4)$$

Then the range-based pulse compression converts the IF signal from time-domain to frequency-domain [9]

$$\begin{aligned} s_{IF}(m, n, f) &= \sum_{p=1}^{N_p} \sigma_p \exp \{ j2\pi f_0 \tau_p \} \\ &\quad \int_0^T \exp \{ j2\pi K \tau_p t \} \times \exp \{ -j2\pi f t \} dt \end{aligned}$$

$$\begin{aligned}
&= \sum_{p=1}^{N_p} \sigma_p \exp\{j2\pi f_0 \tau_p\} \\
&\quad \times T \times \text{sinc}((f - K\tau_p)T) \\
&\quad \times \exp\{-j\pi(f - K\tau_p)T\}
\end{aligned} \quad (5)$$

at a specific range z_p , the frequency $f = K\tau_p$ and the sinc function attains its peak value, the constant T can be ignored and (5) can be represented as [15] and [24]

$$s_{IF}(m, n) = \sum_{p=1}^{N_p} \sigma_p \exp\{j2\pi f_0 \tau_p\}. \quad (6)$$

By collecting all the range-compressed measurement positions, a 2-D beat-frequency slice corresponding to the range z_p is obtained.

2) *Low-Rank Property*: Each range-compressed slice, derived from the EUR sampling in Section II-A, is a structured sparse matrix and must be completed before RMA imaging. Meeting the precondition for MC therefore requires an analysis of the low-rank property. By substituting $\tau_p = 2R_p/c$, (6) can be reformulated as

$$s_{IF}(m, n) = \sum_{p=1}^{N_p} \sigma_p \exp\{j2kR_p\} \quad (7)$$

where $k = 2\pi f_0/c$ refers to the wavenumber. By approximating R_p with (3), (7) can be rewritten as

$$\begin{aligned}
s_{IF}(m, n) &\approx \sum_{p=1}^{N_p} \sigma_p \exp\left\{j2k\left(R_0 + \frac{x_m^2 - 2x_m x_p}{2R_0} + \frac{y_n^2 - 2y_n y_p}{2R_0}\right)\right\} \\
&= \sum_{p=1}^{N_p} \sigma_p \exp\{j2kR_0\} \\
&\quad \times \exp\left\{jk\frac{x_m^2}{R_0}\right\} \times \exp\left\{-j2k\frac{x_m x_p}{R_0}\right\} \\
&\quad \times \exp\left\{-j2k\frac{y_n y_p}{R_0}\right\} \times \exp\left\{jk\frac{y_n^2}{R_0}\right\}.
\end{aligned} \quad (8)$$

When the target size is much smaller than R_0 , the first exponential term in (8) can be ignored as a constant [39], leading to the echo matrix S_{XY} expressed as

$$S_{XY} = D_M X \Gamma Y D_N \in \mathbb{C}^{M \times N} \quad (9)$$

where $D_M \in \mathbb{C}^{M \times M}$ and $D_N \in \mathbb{C}^{N \times N}$ are the diagonal matrices along the two directions, with their elements being $[D_M]_m = \exp\{jk\frac{x_m^2}{R_0}\}$ and $[D_N]_n = \exp\{jk\frac{y_n^2}{R_0}\}$, $\Gamma \in \mathbb{C}^{N_p \times N_p}$ denotes a scattering intensity matrix

$$\Gamma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_{N_p} \end{bmatrix} \quad (10)$$

$X \in \mathbb{C}^{M \times N_p}$ and $Y \in \mathbb{C}^{N_p \times N}$ consist of the steering vectors for scatterers in the vertical and horizontal direction, respectively

$$\begin{aligned}
X &= \begin{bmatrix} X_1^1 & X_1^2 & \cdots & X_1^{N_p} \\ X_2^1 & X_2^2 & \cdots & X_2^{N_p} \\ \vdots & \vdots & \ddots & \vdots \\ X_M^1 & X_M^2 & \cdots & X_M^{N_p} \end{bmatrix}, \\
Y &= \begin{bmatrix} Y_1^1 & Y_1^2 & \cdots & Y_1^{N_p} \\ Y_2^1 & Y_2^2 & \cdots & Y_2^{N_p} \\ \vdots & \vdots & \ddots & \vdots \\ Y_N^1 & Y_N^2 & \cdots & Y_N^{N_p} \end{bmatrix} \\
X_m^p &= \exp\left\{-j2k\frac{x_p}{R_0}x_m\right\}, \quad Y_n^p = \exp\left\{-j2k\frac{y_p}{R_0}y_n\right\}
\end{aligned} \quad (11)$$

when the target is sparse and the scattering points are limited, the matrix Γ exhibits a low-rank property. Since $\text{rank}(S_{XY}) = \text{rank}(D_M X \Gamma Y D_N) \leq \text{rank}(\Gamma) = N_p$, the sparse range slice S_{XY} also has a low-rank property, therefore can be recovered via low-rank MC algorithms.

To summarize, this section presents the system geometry and the EUR scanning strategy. Direct imaging of the EUR-induced structured sparse data would cause severe artifacts, making data completion necessary. By analyzing the echo data, we show that signal sparsity induces a spatial-domain low-rank property, which enables MC. Notably, the above low-rank analysis relies on two assumptions: the sparsity of dominant scattering points, and the far-range approximation used in (3) and (9). The violation or weakening of these two assumptions would degrade the low-rank property of the SAR echoes. The effect of the scattering sparsity is systematically examined in Section IV using targets with different scattering characteristics. The influence of the far-range approximation, along with the effects of scatterer distribution and measurement noise, is investigated through simulation experiments provided in the Supplementary Material.

III. METHODOLOGY

This section first outlines the truncated-DCT for energy concentration and dimensionality reduction, followed by the introduction of Toeplitz matrix methods within the ADMM framework to address structured sparsity and recover missing entries of each range slice. Concretely, the complete workflow of structured sparse mmWave 3-D SAR imaging via truncated-DCT and Toeplitz matrix methods is illustrated in Fig. 4.

A. Truncated-DCT

The discrete cosine transform (DCT), introduced by Ahmed et al. [40], is widely used in image and audio compression as a quasi-optimal transform.

Truncating high-frequency DCT coefficients is a common practice to reduce computational cost, since most

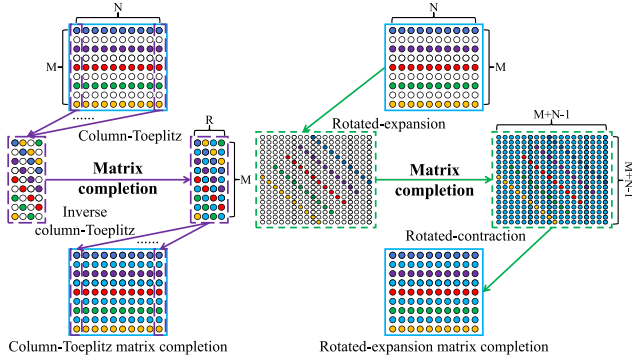


Fig. 6. Column-Toeplitz MC and rotated-expansion MC.

and violates the randomness assumptions in standard MC theory [20]. Established regimes for such structured sparsity include the sequential column-Toeplitz method [39] and the rotated-expansion approach [25]. As shown in Fig. 6, the former is costly and ignores 2-D correlations, while the latter increases dimensionality and introduces structural distortion.

As analyzed in Section II, each range slice represents a superposition of a finite number of complex exponentials, implying an inherent low-rank structure. Toeplitz lifting enhances this property by encoding shift-invariant correlations and redistributes the data into a less structured form to improve incoherence [36], [39]. However, column-wise Toeplitz only exploits single-dimensional correlations and is insufficient to enhance incoherence globally.

In this context, the EUR sampling pattern, characterized by entirely missing rows and fully observed columns, necessitates matrix-level modeling to enable cross-dimensional information propagation. Specifically, the fully sampled column dimension must be leveraged to infer missing rows, which cannot be effectively achieved by purely column-wise methods.

Motivated by these insights, we develop two matrix-level Toeplitz constructions that combine the low-rank enhancement induced by Toeplitz lifting and the cross-dimensional information propagation enabled by matrix-level modeling. This combination improves alignment with the EUR-induced structured missing pattern and enhances MC performance.

The first Toeplitz transformation, termed Toeplitz-Vector (TVec), improves upon the column-wise Toeplitz approach presented in [39] by vectorizing the matrix and applying a single Toeplitz transform. The TVec form of

S_{TD} is $TVec\{S_{TD}\} \in \mathbb{C}^{(ML) \times R}$

$TVec\{S_{TD}\} =$

$$\begin{bmatrix} S_{TD}(1, 1) & S_{TD}(M, L) & \cdots & S_{TD}(M - R + 2, L) \\ S_{TD}(2, 1) & S_{TD}(1, 1) & \cdots & S_{TD}(M - R + 3, L) \\ \vdots & \vdots & \ddots & \vdots \\ S_{TD}(M, L) & S_{TD}(M - 1, L) & \cdots & S_{TD}(M - R + 1, L) \end{bmatrix} \quad (17)$$

while the second one, termed Toeplitz-Column-Toeplitz (TCT), adopts a twofold Toeplitz operation inspired by the Toeplitz covariance structure in radar applications [38]. For TCT, a Toeplitz operation is first conducted

$TCT\{S_{TD}(:, l)\} =$

$$\begin{bmatrix} S_{TD}(1, l) & S_{TD}(M, l) & \cdots & S_{TD}(M - P + 2, l) \\ S_{TD}(2, l) & S_{TD}(1, l) & \cdots & S_{TD}(M - P + 3, l) \\ \vdots & \vdots & \ddots & \vdots \\ S_{TD}(M, l) & S_{TD}(M - 1, l) & \cdots & S_{TD}(M - P + 1, l) \end{bmatrix} \quad (18)$$

$TCT\{S_{TD}(:, l)\} \in \mathbb{C}^{M \times P}$ refers to the submatrix of column l and L columns correspond to L submatrices, which are arranged in a column and then subjected to a second Toeplitz operation to obtain $TCT\{S_{TD}\} \in \mathbb{C}^{(ML) \times (PQ)}$ in (19) shown at the bottom of this page.

R in (17) is the pencil parameter for TVec, P, Q in (18) and (19) are the pencil parameters for TCT. The two Toeplitz operations are more intuitively demonstrated in Fig. 7.

C. Matrix Completion

The Toeplitz matrices with enhanced incoherence are described in Section III-B, leading to superior MC performance. To complete the compressed echo slice, this subsection introduces a class of Toeplitz matrix methods based on the SVD-free MC algorithm [44]. These methods fuse the proposed TVec/TCT matrix within the factorized ADMM framework, namely TVecA/TCTA.

Let $\hat{S}_{TD}$ denote the corresponding sparse matrix to be recovered. For convenience, we use $T\{S_{TD}\}$ to represent the Toeplitz matrix of S_{TD} in either the TVec or TCT form. Similarly, $T\{\hat{S}_{TD}\}$ is used to represent the Toeplitz matrices of $\hat{S}_{TD}$. Then the optimization model can be established as

$$\begin{aligned} \min \quad & Rank(T\{\hat{S}_{TD}\}) \\ \text{s.t.} \quad & P_{\Omega}(\hat{S}_{TD}) = P_{\Omega}(S_{TD}) \end{aligned} \quad (20)$$

$$TCT\{S_{TD}\} = \begin{bmatrix} TCT\{S_{TD}(:, 1)\} & TCT\{S_{TD}(:, L)\} & \cdots & TCT\{S_{TD}(:, L - Q + 2)\} \\ TCT\{S_{TD}(:, 2)\} & TCT\{S_{TD}(:, 1)\} & \cdots & TCT\{S_{TD}(:, L - Q + 3)\} \\ \vdots & \vdots & \ddots & \vdots \\ TCT\{S_{TD}(:, L)\} & TCT\{S_{TD}(:, L - 1)\} & \cdots & TCT\{S_{TD}(:, L - Q + 1)\} \end{bmatrix}_{(ML) \times (PQ)}. \quad (19)$$

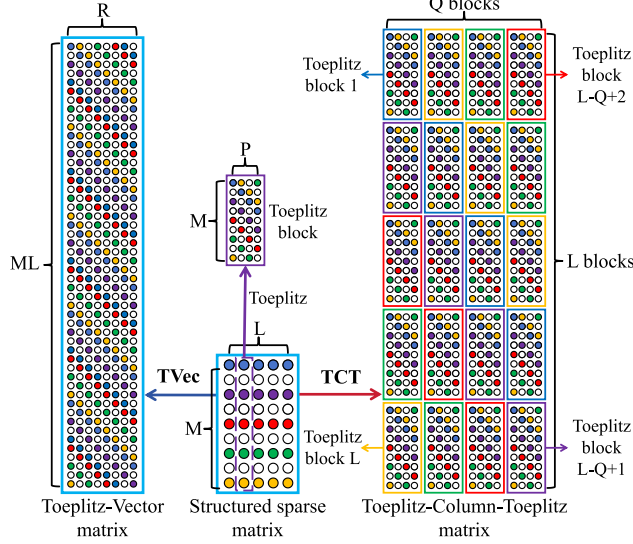


Fig. 7. Toeplitz-Vector matrix and TCT matrix.

where Ω denotes the sampling entry mask of the sparse matrix S_{TD} . $P_{\Omega}(\cdot)$ is the projection mapping on Ω . Conventionally, nuclear norm minimization is used as convex relaxation, which is given by

$$\begin{aligned} \min \quad & \|T\{\hat{S}_{TD}\}\|_* \\ \text{s.t.} \quad & P_{\Omega}(\hat{S}_{TD}) = P_{\Omega}(S_{TD}). \end{aligned} \quad (21)$$

Traditional approaches to nuclear norm minimization rely on SVD, which exhibits $O(M^2N)$ computational complexity for a matrix of $M \times N$ dimension. This complexity renders SVD-based methods infeasible for the large-scale expanded Toeplitz matrices. In order to reduce the computational load, the expanded TVec/TCT matrix is factorized into two disjoint components [45], [46], and (21) is then restated as

$$\begin{aligned} \min \quad & \|U\|_F^2 + \|V\|_F^2 \\ \text{s.t.} \quad & P_{\Omega}(\hat{S}_{TD}) = P_{\Omega}(S_{TD}), T\{\hat{S}_{TD}\} = UV^H \end{aligned} \quad (22)$$

where $\|\cdot\|_F$ is the Frobenius norm, $T\{\hat{S}_{TD}\} = UV^H$ is the factorization of $T\{\hat{S}_{TD}\}$, and $[\cdot]^H$ denotes the conjugate transpose of the matrix. Then the ADMM is leveraged to effectively solve (22), which avoids SVD operations and speeds up iterations. The augmented Lagrangian function is defined as [44]

$$\begin{aligned} L(\hat{S}_{TD}, U, V, Z) = & \text{Ind}_C(\hat{S}_{TD}) + \frac{1}{2} (\|U\|_F^2 + \|V\|_F^2) \\ & + \langle Z, T\{\hat{S}_{TD}\} - UV^H \rangle \\ & + \frac{\mu}{2} \|T\{\hat{S}_{TD}\} - UV^H\|_F^2 \end{aligned} \quad (23)$$

where μ is the penalty parameter, Z is the Lagrange multiplier, $\text{Ind}_C(\hat{S}_{TD})$ is the indicator function defined as

$$\text{Ind}_C(\hat{S}_{TD}) = \begin{cases} 0, & P_{\Omega}(S_{TD}) = P_{\Omega}(\hat{S}_{TD}) \\ \infty, & \text{otherwise.} \end{cases} \quad (24)$$

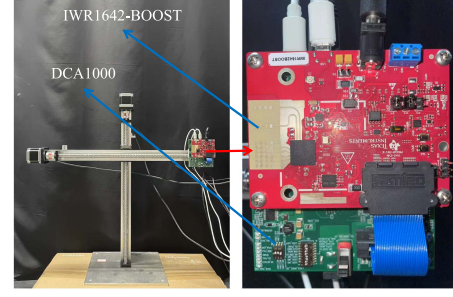


Fig. 8. Experimental mmWave 3-D SAR imaging system.

One of the advantages of ADMM is that each subproblem can be obtained from (23). Specifically, we have

$$\begin{cases} \hat{S}_{TD}^{(k+1)} = \arg \min_{\hat{S}_{TD}} \{L(\hat{S}_{TD}, U^{(k)}, V^{(k)}, Z^{(k)})\} \\ U^{(k+1)} = \arg \min_U \{L(\hat{S}_{TD}^{(k+1)}, U, V^{(k)}, Z^{(k)})\} \\ V^{(k+1)} = \arg \min_V \{L(\hat{S}_{TD}^{(k+1)}, U^{(k+1)}, V, Z^{(k)})\} \\ Z^{(k+1)} = Z^{(k)} + \mu(T\{\hat{S}_{TD}^{(k+1)}\} - U^{(k+1)}V^{(k+1)H}) \end{cases} \quad (25)$$

by setting $\partial L / \partial \hat{S}_{TD} = 0$, the update method for $\hat{S}_{TD}^{(k+1)}$ can be obtained as

$$\begin{aligned} \hat{S}_{TD}^{(k+1)} = & P_{\Omega}(S_{TD}) + \\ & P_{\Omega^c} \left(T^+ \left\{ U^{(k)}V^{(k)H} - \frac{1}{\mu} Z^{(k)} \right\} \right) \end{aligned} \quad (26)$$

here, Ω^c denotes the missing entry mask. T^+ is the Moore-Penrose pseudo-inverse mapping from Toeplitz matrix to the original one, which calculates the average of the corresponding elements and maps them back to the coordinates of the original matrix [32].

Likewise, by letting $\partial L / \partial U = 0$ and $\partial L / \partial V = 0$, we obtain

$$\begin{aligned} U(I + \mu V^H V) &= (\mu T\{\hat{S}_{TD}\} + Z)V \\ V(I + \mu U^H U) &= (\mu T\{\hat{S}_{TD}\} + Z)^H U \end{aligned} \quad (27)$$

here I is the identity matrix, then the solution of U can be expressed as

$$\begin{aligned} U^{(k+1)} &= (\mu T\{\hat{S}_{TD}^{(k+1)}\} + Z^{(k)})V^{(k)} \\ & \quad (I + \mu V^{(k)H} V^{(k)})^{-1} \end{aligned} \quad (28)$$

and V is then updated by

$$\begin{aligned} V^{(k+1)} &= (\mu T\{\hat{S}_{TD}^{(k+1)}\} + Z^{(k)})^H U^{(k+1)} \\ & \quad (I + \mu U^{(k+1)H} U^{(k+1)})^{-1}. \end{aligned} \quad (29)$$

Despite the large size of $T\{\hat{S}_{TD}\}$, its factorization into U and V reduces the computational burden to inverting only an $r_e \times r_e$ matrix [32], where r_e denotes the estimated rank of the Toeplitz matrices. Using another SVD-free algorithm, alternating least squares (ALS) [47] for the initial factorization of $T\{\hat{S}_{TD}\}$ and ADMM for optimization, TVecA/TCTA converges within tens of iterations [34], reaching a precision suitable for sparse SAR imaging.

Once the compressed echo matrix $\hat{S}_{TD} \in \mathbb{C}^{M \times L}$ has been recovered through TVecA/TCTA, the completed matrix $\hat{S}_{XY} \in \mathbb{C}^{M \times N}$ corresponding to the sparse range slice S_{XY}

Algorithm 1 TD-TVecA/TCTA**Require:**A sparse range slice matrix: $S_{XY} \in \mathbb{C}^{M \times N}$ **Output:**The reconstructed range slice matrix: $\hat{S}_{XY} \in \mathbb{C}^{M \times N}$

1: Initialization: Set the truncation threshold

 $L = N/2$, the number of iterations $K = 30$, the iteration $k = 0$, the penalty parameter $\mu = 2$, the Lagrange multiplier $Z^{(0)} = 0$.2: Perform truncated-DCT on S_{XY} :

$$\hat{S}_{TD}^{(0)} = S_{TD} = IDCT(DCT(S_{XY})_{(:,1:L)})$$

3: Factorize $T\{\hat{S}_{TD}^{(0)}\}$ into $U^{(0)}$ and $V^{(0)}$ using ALS:

$$U^{(0)}V^{(0)H} = T\{\hat{S}_{TD}^{(0)}\}$$

4: **while** $k \leq K$ **do**

$$5: \quad \hat{S}_{TD}^{(k+1)} = P_{\Omega}(S_{TD}) + P_{\Omega^c}(T^+\{U^{(k)}V^{(k)H} - \frac{1}{\mu}Z^{(k)}\})$$

$$6: \quad U^{(k+1)} = (\mu T\{\hat{S}_{TD}^{(k+1)}\} + Z^{(k)})V^{(k)}(I + \mu V^{(k)H}V^{(k)})^{-1}$$

$$7: \quad V^{(k+1)} = (\mu T\{\hat{S}_{TD}^{(k+1)}\} + Z^{(k)})^H U^{(k+1)}(I + \mu U^{(k+1)H}U^{(k+1)})^{-1}$$

$$8: \quad Z^{(k+1)} = Z^{(k)} + \mu(T\{\hat{S}_{TD}^{(k+1)}\} - U^{(k+1)}V^{(k+1)H})$$

9: $k = k + 1$ 10: **end while**11: Perform inverse truncated-DCT on $\hat{S}_{TD}^{(K)}$:

$$\hat{S}_{XY} = IDCT\left(\left[DCT(\hat{S}_{TD}^{(K)}) \mid 0\right]\right)$$

12: **return** $\hat{S}_{XY} \in \mathbb{C}^{M \times N}$

can be derived by the inverse truncated-DCT shown in Fig. 5. As an example, Algorithm 1 presents the workflow of the TD-TVecA/TCTA algorithm, incorporating the truncated-DCT and Toeplitz matrix methods.

D. Computational Complexity

The computational complexity of the proposed TD-TVecA/TCTA mainly stems from the updates of $\hat{S}_{TD}$, U , V , and Z .

For the compressed matrix $S_{TD} \in \mathbb{C}^{M \times L}$, the onefold TVec transformation costs $O(MLR)$, whereas the computational complexity of the twofold TCT is $O(LMP + MLPQ)$. Despite TVec or TCT, factorizing $T\{S_{TD}\}$ for both expanded matrices requires $O(MLr_e^2)$. Since T^+ represents the pseudo-inverse of TVec/TCT, updating $\hat{S}_{TD}$ and Z has the same complexity as Toeplitz transformations. Similarly, updating U and V is dominated by matrix multiplication, resulting in a complexity of $O(MLRr_e)$ or $O(MLPQr_e)$, while the accompanying $O(r_e^3)$ matrix inversion is negligible.

Taking the iteration cycles into account, the computational burden of recovering a range slice via TD-TVecA/TCTA is mainly in the order of $O(MLr_e^2 + KMLRr_e)$ or $O(MLr_e^2 + KMLPQr_e)$. The TD-TVecA, benefiting from the lower complexity, allows for faster completion. In contrast, the TD-TCTA exploits the holistic structure of the data more effectively with a higher computational cost, leading to superior reconstruction performance

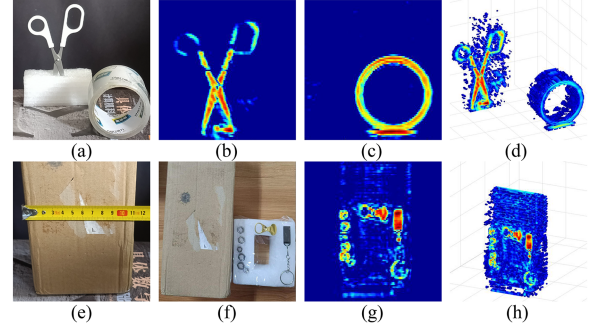


Fig. 9. Imaging results under full sampling. (a) Optical image of scissors and adhesive tape. (b) 2-D imaging result of scissors at 40 cm. (c) 2-D imaging result of adhesive tape at 25 cm. (d) 3-D imaging result of the complete scene. (e) Optical image and size measurement of a box. (f) Optical image of various objects on foam. (g) 2-D tomographic imaging result of objects inside the box. (h) 3-D tomographic imaging result of objects inside the box.

TABLE I
Parameter Settings of Physical Imaging System

Parameter	Value
Signal Type	FMCW
Center Frequency	79 GHz
Frequency Slope	63.343 MHz/ μ s
Bandwidth	4 GHz
Sampling Rate	5.12 MHz
ADC sample points	256
Pulse Repetition Frequency (PRF)	20 Hz
Radar Moving Speed	30 mm/s
Virtual Aperture($Dx \times Dy$)	198 mm $\times$ 300 mm
Sampling Interval($dx \times dy$)	2 mm $\times$ 1.5 mm
Sampling Points($M \times N$)	100 $\times$ 201

in highly sparse scenarios. Experimental comparisons are provided in Section IV.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This section first physically implements the mmWave SAR imaging system and validates the effectiveness of the system under full sampling. Subsequently, comparative experiments are conducted on two types of targets with different scattering characteristics.

A. Experimental Setup and Full Sampling Results

In this article, a mmWave 3-D SAR imaging system is deployed to realize the geometry shown in Fig. 1. The experimental setup, illustrated in Fig. 8, consists of a TI IWR1642-Boost radar sensor, a DCA1000 data acquisition card, a two-axis translation stage, and a PC for sensor control, motion control, and data processing.

The IWR1642 radar employs 2 transmit and 4 receive antennas. All antennas are vertically polarized, and Tx1 and Rx1 are selected to form a bistatic configuration. The radar is integrated with the DCA1000 and mounted on a 2-D scanning platform equipped with two orthogonal linear actuators. All experiments are conducted on a PC with Intel Core i7-7700HQ 2.80 GHz CPU and 16 GB RAM, using MATLAB R2018a. The parameter settings of the system are detailed in Table I. For all subsequent imaging results, the

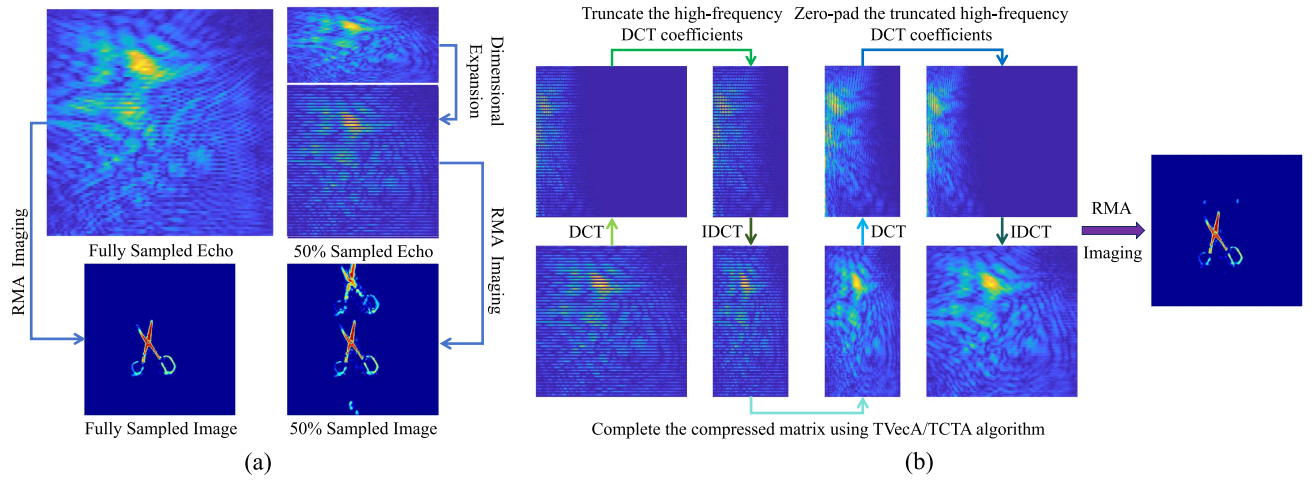


Fig. 10. Illustration of echo recovery and imaging. (a) Echo data and corresponding images for the fully sampled case and the 50% EUR sampled case. The sparse data on the right results from uniform-interval sampling followed by row-wise dimensional expansion via interpolation. (b) Complete workflow of recovering 50% EUR sampled echo data using TD-TVecA/TCTA and the resulting image after data completion.

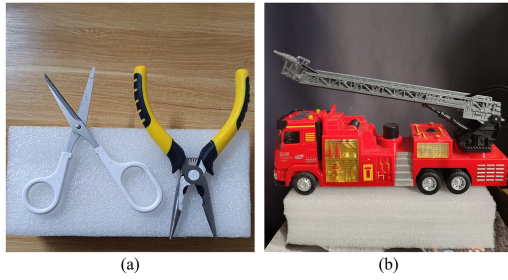


Fig. 11. Optical images of targets. (a) Target 1: Scissors and Pliers. (b) Target 2: Fire Truck Model.

TABLE II
Average Runtime Comparison (Mean $\pm$ Standard Deviation)

Algorithms	TSVT	TIALM	TTSPN-SR	Proposed TD-TVecA
Time (s)	329.2 $\pm$ 17.6	10.3 $\pm$ 0.9	159.4 $\pm$ 11.4	2.5 $\pm$ 0.2
Algorithms	RSVT	RIALM	RTSPN-SR	Proposed TD-TCTA
Time (s)	57.6 $\pm$ 3.7	2.9 $\pm$ 0.2	29.8 $\pm$ 2.5	3.1 $\pm$ 0.2

amplitudes of scatterers are normalized to their maximum and shown on a $[-40, 0]$ dB scale.

To verify the effectiveness and penetrability of the mmWave 3-D SAR imaging system under full sampling, scissors, an adhesive tape, nuts, a can pull-tab, and a USB drive are imaged at a range of approximately 20–50 cm. Fig. 9 demonstrates the imaging results of these items. The scissors and the tape measure approximately 10–15 cm in size, with the tape having a thickness of 6 cm. They are positioned at distances of 25 and 40 cm, respectively. Fig. 9(a) displays the optical image and their 2-D and 3-D images are shown in Fig. 9(b)–(d). Fig. 9(e) shows a box containing an 11 cm wide foam block that holds multiple objects, including nuts with inner diameters of 0.7 and 0.5 cm, a pull-tab and a USB drive. Fig. 9(f) shows their detailed layout. Corresponding 2-D and 3-D tomographic imaging results are presented in Fig. 9(g) and (h), respectively.

It can be observed that the full sampling mode demonstrates excellent capabilities in 3-D tomographic imaging. Fig. 9(g) indicates that the actual resolution of the imaging system is approximately 5–7 mm, which is sufficient for practical detection purposes. Fig. 9(d) and 9(h) shows that the system enables penetrating 3-D imaging at varying distances, allowing for the precise detection of concealed objects.

However, despite the time of processing the collected data and performing 3-D RMA imaging only costs a few tens of seconds, the full sampling scan takes 1000 s (approximately 17 min), which is evidently impractical for real-world applications.

B. Structured Sparse Sampling Results

To accelerate the practical scanning process, 50% and 25% EUR sampling modes are considered. Under these configurations, the proposed TD-TVecA/TCTA is compared with SVT [26], IALM [28], and TSPN-SR [39], each combined with column-Toeplitz [39] and rotated-expansion [25] preprocessing, yielding TSVT, RSVT, TIALM, RIALM, TTSPN-SR, and RTSPN-SR.

For the proposed TD-TVecA/TCTA, we set the estimated rank $r_e = 20$, and the pencil parameters $R = 4 \times r_e$, $P = Q = \lfloor r_e/2 \rfloor$. For other algorithms, the parameter settings follow the original papers.

Fig. 10 details the TD-TVecA/TCTA-based completion and imaging of 50% EUR sampled echo for the scissors as an example. The fully sampled echo is compared against 50% sampled data, whose direct reconstruction shows severe periodic sidelobes in Fig. 10(a). In contrast, the TD-TVecA/TCTA recovery in Fig. 10(b) achieves precise echo completion and a final image with excellent sidelobe suppression.

Following the validation on a single object, the effectiveness of the proposed methods is further evaluated using two distinct imaging targets with varying scattering characteristics, as shown in Fig. 11. Target 1 comprises a

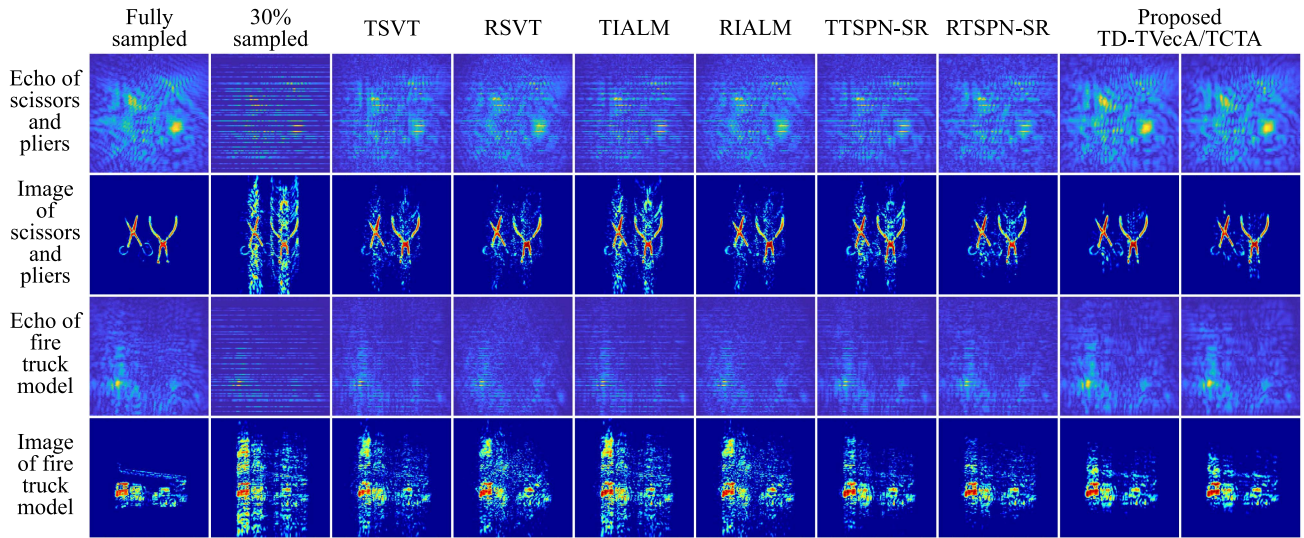


Fig. 12. Echoes and imaging results of targets under 30% non-uniform row undersampling. Rows 1 and 2 correspond to the echoes and imaging results of Target 1, respectively. Similarly, Rows 3 and 4 are for Target 2. Columns 1 and 2 display the fully sampled echo and its direct imaging result, and the 30% sampled echo and its direct imaging result, respectively. The remaining columns represent the echo completion effects of different algorithms and the imaging results using the completed echoes.

pair of scissors and pliers, representing sparse scatterers with limited details, both measuring 10 – 15 cm. Target 2 is a 26 cm $\times$ 9 cm $\times$ 14 cm fire truck model with numerous dense scatterers and complex details.

C. 30% Nonuniform Row Undersampling

First, the computational burden of different algorithms is evaluated via Monte Carlo experiments under 30% non-uniform row undersampling. For each of the targets, 50 random non-uniform sampling sets were generated and processed, yielding 100 trials in total. The average time consumption is summarized in Table II, while the reconstruction results of one representative set are shown in Fig. 12.

As observed from Table II and Fig. 12, SVT, IALM, and TSPN-SR, when augmented by either the column-Toeplitz or rotated-expansion strategy, can all achieve a certain level of completion effectiveness. Among them, the rotated-expansion requires approximately one-fifth of the time of the column-Toeplitz while yielding superior reconstruction performance.

Notably, the fast-converging [28] RIALM method achieves acceptable imaging results in just 2.9 s on average. Meanwhile, RTSPN-SR, despite a higher computational time of 29.8 s, produces imaging results comparable to the proposed methods.

Finally, it is found that both proposed methods can achieve the optimal echo reconstruction and imaging results within 5 s. The imaging results of TD-TVecA and TD-TCTA for the two targets have their own merits, with TD-TCTA requiring a longer average time. This is consistent with the computational load analyzed in Section III-D. In subsequent experiments, the average time of different algorithms is generally consistent with the results shown in Table II.

D. 50% and 25% EUR sampling

In contrast to nonuniform row undersampling, the 50% and 25% EUR sampling can reduce the required scanning time practically from 17 min to approximately 8 min and 4 min, respectively.

We perform reconstruction and imaging under these two more structured sparse patterns and present the corresponding results in Fig. 13. The echo reconstruction is evaluated using the structural similarity index measure (SSIM) and peak signal-to-noise ratio (PSNR), while the imaging quality is assessed by correlation coefficient (CORR) and root-mean-square error (RMSE), as summarized in Tables III and IV.

Fig. 13(a) reveals that, in contrast to non-uniform sampling, SVT and IALM prove largely ineffective for reconstruction under EUR sampling. TSVT and RSVT perform worst, yielding results on par with or worse than direct imaging from sparse data. TSPN-SR, however, maintains its effectiveness as seen previously. Again, RTSPN-SR outperforms TTSPN-SR, delivering image quality closer to ours, though still inferior in echo completion.

For the 25% EUR sampling in Fig. 13(b), the proposed methods retain strong recovery capabilities, especially in suppressing sidelobes for Target 1. While RTSPN-SR's performance degrades further, our methods successfully recover all missing rows of echoes.

Overall, TD-TVecA and TD-TCTA excel at completing echoes in high-energy regions, with a slight performance drop in low-energy areas. This limitation originates from the high-frequency coefficients loss by truncated-DCT. Nevertheless, this loss is acceptable given the associated halving of computational time. Further evaluation of the truncated-DCT is provided in Section IV-E

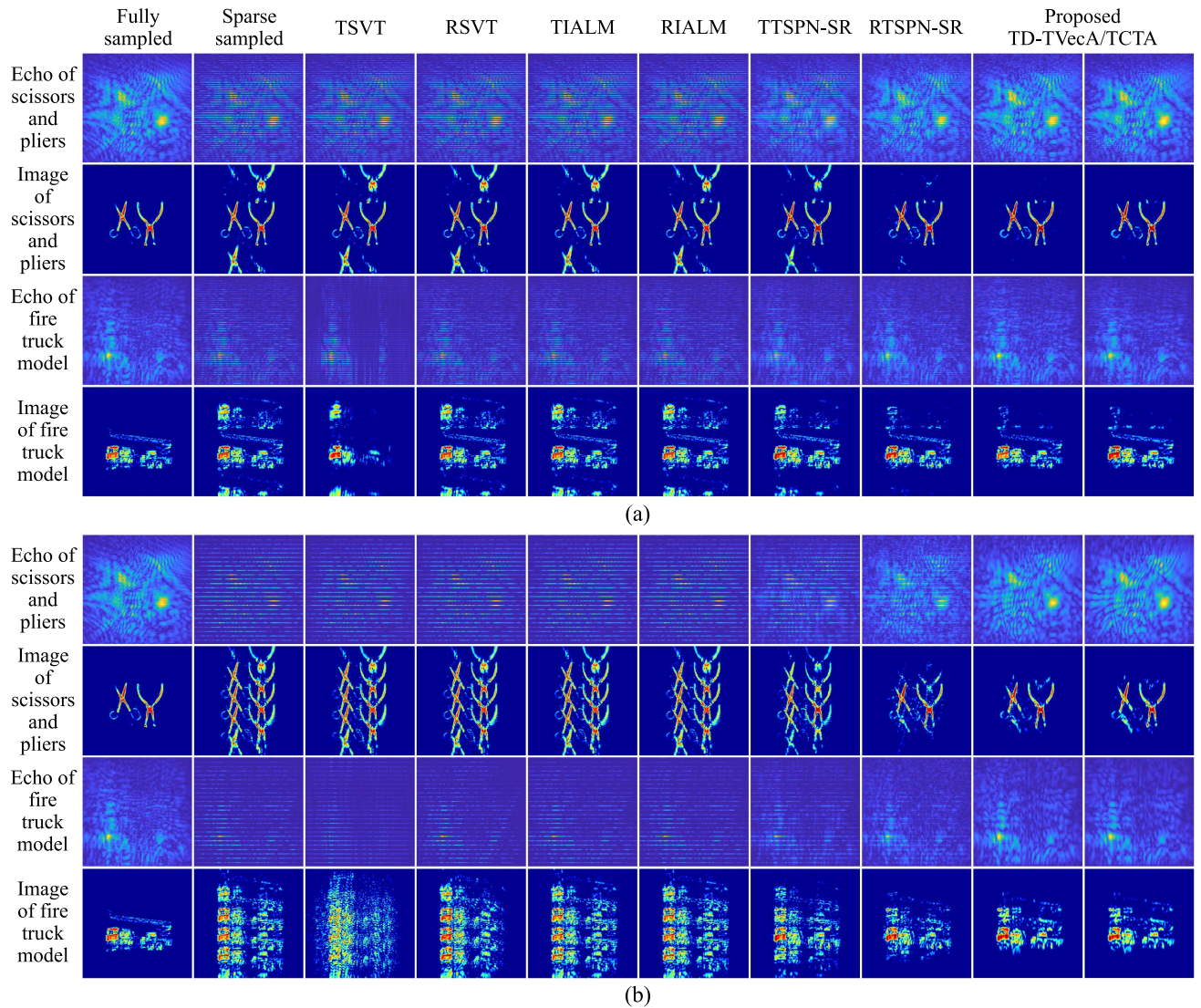


Fig. 13. Echoes and imaging results of two targets under 50% and 25% EUR sampling. The odd-numbered rows display the echoes of the imaging targets, while the even-numbered rows show the corresponding imaging results. The first two columns present the direct imaging results of the fully sampled and the undersampled data. The other columns display the results after applying different completion algorithms to the sparse echoes, followed by RMA imaging. (a) 50% EUR sampling. (b) 25% EUR sampling.

The proposed algorithms establish a new state-of-the-art for reconstructing SAR data under 50% and 25% EUR sampling mode, as evidenced by the comparative results in Tables III and IV, respectively.

Metrics indicate that SVT and IALM methods fail entirely, and TSPN-SR methods are effective. While ours consistently rank first or second in all metrics, showing a particularly strong lead for Target 1 (SSIM of TD-TCTA = 0.7111, SSIM of RTSPN-SR = 0.5849; PSNR of TD-TCTA = 26.20dB, PSNR of RTSPN-SR = 23.53 dB).

A comparison between Tables III and IV reveals that, as the sampling rate decreases, the proposed methods consistently maintain a clear advantage over RTSPN-SR for both Target 1 and Target 2. For Target 1, When the sampling rate is reduced from 50% to 25%, the CORR difference between ours and RTSPN-SR increases from 0.0095 (CORR of TD-TCTA = 0.9818, CORR of RTSPN-SR =

0.9723) to 0.0513 (CORR of TD-TCTA = 0.9091, CORR of RTSPN-SR = 0.8578). For Target 2, the CORR difference decreases slightly from 0.0107 to 0.0053, indicating that the performance gap narrows as the scene becomes more complex.

In summary, the proposed methods remain robust across diverse targets and sampling rates. TD-TVecA is computationally more efficient due to its simpler transformation, whereas TD-TCTA demonstrates superior performance under severe sparsity (25% EUR sampling) by more effectively exploiting 2-D structural information. In addition, noise robustness analysis (see Supplementary Material) shows that the proposed method maintains stable reconstruction performance up to a signal-to-noise ratio of 20 dB under EUR sampling.

These experimental results also help clarify the applicability of the proposed methods. The underlying low-rank

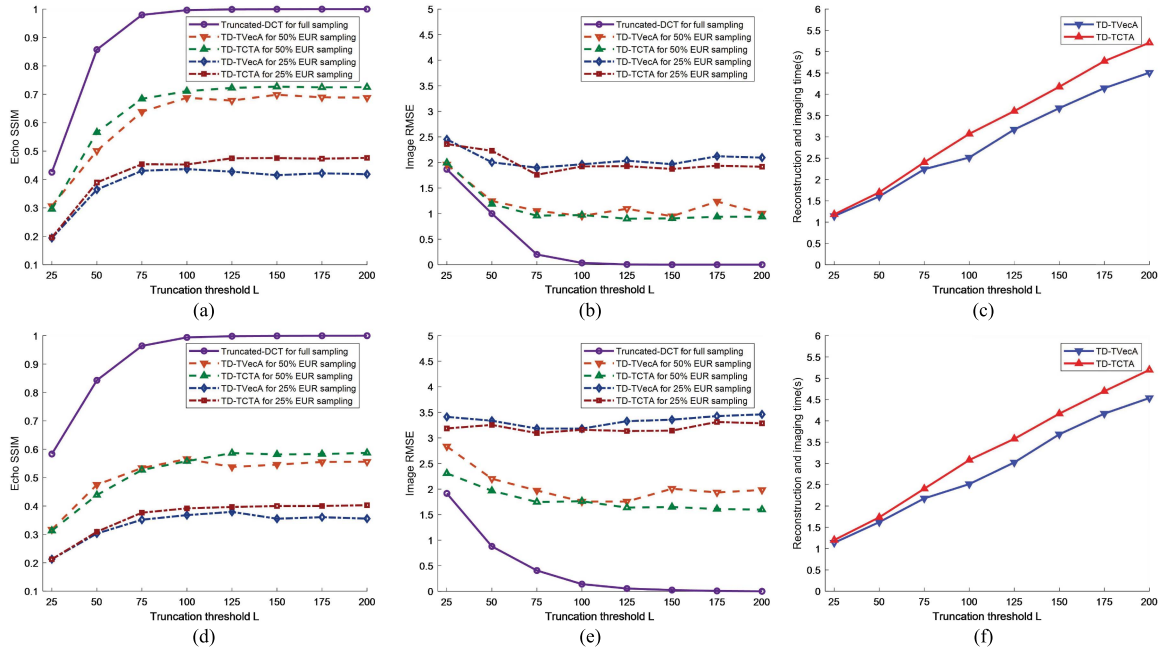


Fig. 14. Performance of TD-TVecA and TD-TCTA under 50% and 25% EUR sampling with varying truncation threshold L for Target 1 and Target 2 in terms of echo SSIM, image RMSE, and running time. (a) Echo SSIM of Target 1. (b) Image RMSE of Target 1. (c) Running time of Target 1. (d) Echo SSIM of Target 2. (e) Image RMSE of Target 2. (f) Running time of Target 2.

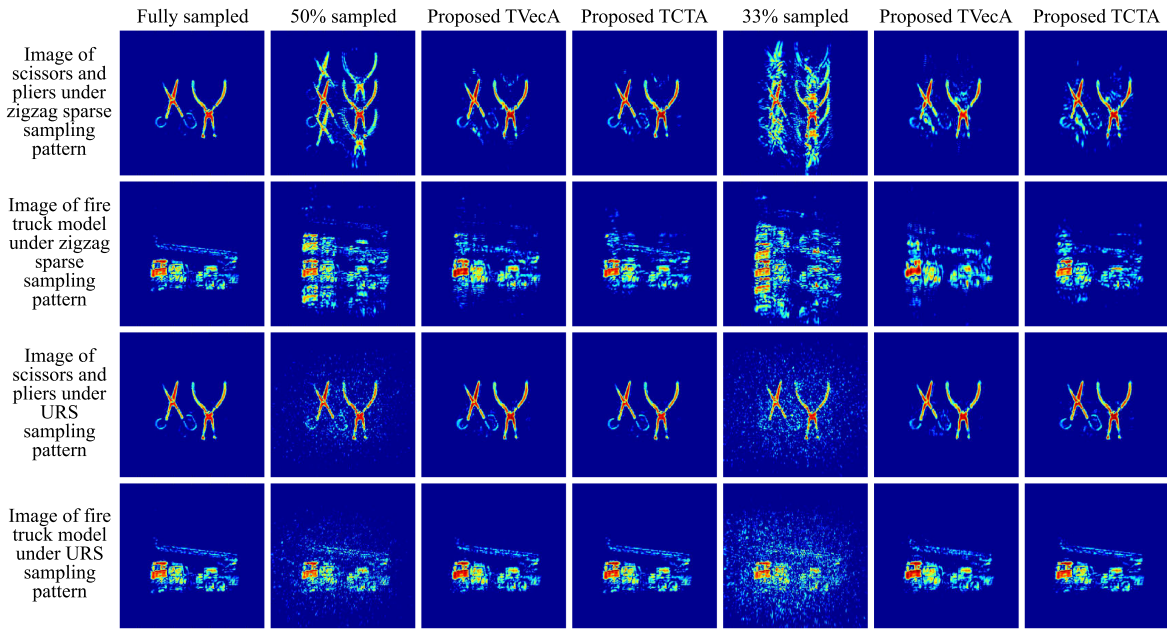


Fig. 15. Imaging results of two targets under zigzag and URS sampling at rates of 50% and 33%. Rows 1 and 2 are the imaging results under zigzag sampling, and rows 3 and 4 are under URS sampling. Column 1 presents the fully sampled imaging results. Columns 2 and 5 show the sparse imaging results. Columns 3 and 6 present imaging results by TVecA. Columns 4 and 7 display the imaging results by TCTA.

assumption is most valid for scenes dominated by a limited number of isolated or weakly clustered scatterers (Target 1), where each range slice can be well represented by a small set of spatial frequency components. For more complex scenes involving dense scatterer distributions or extended targets, the number of effective spatial frequencies increases, leading to higher rank structures and a corresponding degradation in reconstruction performance, as observed for Target 2.

Finally, it should be noted that the comparative experiments presented in this subsection are not intended to provide an exhaustive evaluation of all structured low-rank reconstruction approaches. Instead, they aim to highlight the benefits of explicitly exploiting shift-invariant Toeplitz structures under structured row-wise undersampling. Future work will consider more comprehensive comparisons with additional structured low-rank reconstruction methods.

TABLE III
Performance Metrics of Completing and Imaging Under 50% EUR Sampling Mode

Object	Metric	Fully sampled	50% sampled	TSVT	RSVT	TIALM	RIALM	TTSPN -SR	RTSPN -SR	Proposed TD-TVecA/TCTA
Echo of scissors and pliers	SSIM $\uparrow$	1	0.2324	0.2204	0.2322	0.2325	0.2324	0.3828	0.5849	0.6878/ 0.7111
	PSNR $\uparrow$	∞	15.5450	15.5432	15.5434	15.5450	15.5450	17.9694	23.5268	25.6924/ 26.2017
Image of scissors and pliers	CORR $\uparrow$	1	0.7386	0.7371	0.7387	0.7385	0.7386	0.8360	0.9723	0.9812/ 0.9818
	RMSE $\downarrow$	0	3.6660	3.7260	3.6643	3.6660	3.6660	2.8219	1.0804	0.9547 /0.9716
Echo of fire truck model	SSIM $\uparrow$	1	0.2561	0.0322	0.2426	0.2561	0.2561	0.4281	0.5386	0.5657 /0.5580
	PSNR $\uparrow$	∞	20.1197	19.1891	20.0975	20.1197	20.1197	22.9364	26.2902	28.3089 /28.2673
Image of fire truck model	CORR $\uparrow$	1	0.7500	0.6070	0.7500	0.7500	0.7500	0.8554	0.9240	0.9285/ 0.9347
	RMSE $\downarrow$	0	3.8668	4.0019	3.8668	3.8668	3.8668	2.9218	1.8692	1.7557 /1.7618

$\uparrow$ indicates that a higher value is better, while $\downarrow$ denotes that a lower value is preferred. The optimal values are highlighted in bold.

TABLE IV
Performance Metrics of Completing and Imaging Under 25% EUR Sampling Mode

Object	Metric	Fully sampled	25% sampled	TSVT	RSVT	TIALM	RIALM	TTSPN -SR	RTSPN -SR	Proposed TD-TVecA/TCTA
Echo of scissors and pliers	SSIM $\uparrow$	1	0.0851	0.0697	0.0843	0.0851	0.0851	0.1666	0.3135	0.4366/ 0.4529
	PSNR $\uparrow$	∞	13.7776	13.7701	13.7734	13.7776	13.7776	15.3569	19.6679	22.5322 /22.4164
Image of scissors and pliers	CORR $\uparrow$	1	0.5038	0.5011	0.5041	0.5038	0.5038	0.6302	0.8578	0.9042/ 0.9091
	RMSE $\downarrow$	0	6.3751	6.3731	6.3151	6.3751	6.3751	4.7773	2.4597	1.9637/ 1.9226
Echo of fire truck model	SSIM $\uparrow$	1	0.0895	0.0086	0.0466	0.0895	0.0895	0.1912	0.3012	0.3680/ 0.3919
	PSNR $\uparrow$	∞	18.3311	17.9346	18.1918	18.3311	18.3311	20.3031	23.3037	25.7718/ 26.1471
Image of fire truck model	CORR $\uparrow$	1	0.5043	0.3799	0.4657	0.5043	0.5043	0.6657	0.7808	0.7830/ 0.7861
	RMSE $\downarrow$	0	6.9486	7.5600	7.3198	6.9486	6.9486	4.8753	3.2897	3.1827/ 3.1596

$\uparrow$ indicates that a higher value is better, while $\downarrow$ denotes that a lower value is preferred. The optimal values are highlighted in bold.

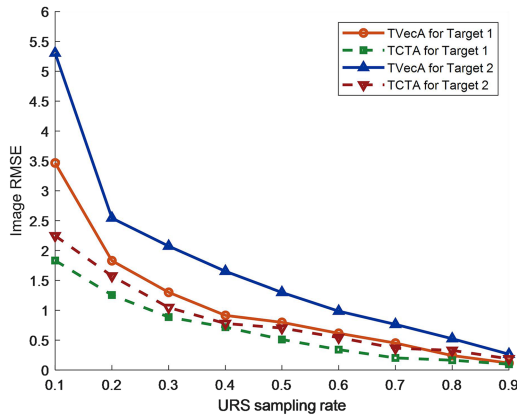


Fig. 16. Performance of TVecA and TCTA with varying URS sampling rates for Targets 1 and Target 2 in terms of Image RMSE.

E. Ablation Studies

This section first examines the efficiency and trade-offs of incorporating truncated-DCT into Toeplitz matrix methods for reconstruction. Subsequently, we present the generality of Toeplitz matrix methods under different sparse sampling patterns.

1) *Efficiency of Truncated-DCT*: As discussed in Section III-A, truncated-DCT introduces a controlled approximation by discarding high-frequency components. This subsection quantitatively evaluates its impact under full sampling, 50% and 25% EUR sampling for both Target 1 and Target 2 by varying the truncation threshold L .

Fig. 14 illustrates the influence of L on the proposed TD-TVecA and TD-TCTA algorithms, where L is smaller than the row dimension of the echo matrix ($N = 201$). Echo completion is quantified using SSIM, while imaging quality is measured by RMSE. Additionally, the effect of L on reconstruction and imaging time is also reported. All results are averaged over ten independent trials.

From Fig. 14(a) and (b), it can be observed that lower sampling rates lead to reduced variations in algorithm performance with respect to the truncation threshold L . In the case of full sampling, when the truncation threshold L exceeds 100 ($\approx N/2$), the truncated-DCT provides negligible improvement in echo and imaging quality. This indicates that the echo matrix can be reliably compressed to half its original dimension while maintaining reconstruction fidelity. Comparing Fig. 14(a) with (d) and Fig. 14(b) with (e), it is evident that the algorithms achieve better performance and exhibit greater sensitivity to changes in L for the simpler Target 1. Fig. 14(c) and (f) further shows that the runtime trends of both TD-TVecA and TD-TCTA are similar across different targets and scale nearly linearly with L , while TD-TVecA remains consistently faster.

As shown in Fig. 14(a)–(e), both TD-TVecA and TD-TCTA consistently achieve optimal or near-optimal performance at $L = N/2$ across different targets and sparsity levels. When L exceeds $N/2$, performance improvement saturates, whereas reconstruction and imaging times continue to grow. At $L = 200 (\approx N)$, TD-TVecA and TD-TCTA degenerate to the nontruncated versions, TVecA and TCTA.

Despite a marginal gain in performance compared with $L = N/2$, Fig. 14(c) and (f) indicates that TVecA and TCTA ($L = N$) demand nearly twice the computational time.

These results demonstrate that the incorporation of truncated-DCT for both TVecA/TCTA not only reduces the runtime by nearly 50%, but also preserves echo reconstruction quality and imaging performance. This enables the proposed algorithm to achieve near real-time reconstruction in practical scenarios.

2) *Generality of Toeplitz Matrix Methods:* A further evaluation is conducted to verify the general applicability of the proposed Toeplitz matrix methods. We perform reconstruction and imaging for Target 1 and 2 using TVecA and TCTA under zigzag [39] and URS sampling patterns. The sampling rates are set to 50% and 33%. The corresponding results are presented in Fig. 15.

As shown in Fig. 15, the proposed algorithm maintains excellent performance across diverse sampling schemes. These results demonstrate their robustness and generalization ability. Notably, the imaging results with zigzag sampling are inferior to those with URS when operating at the same sampling rate, since the zigzag pattern also introduces structured sparsity, though less extreme than that generated by EUR sampling.

From the visual results, it is evident that for Target 2 under 33% URS sampling, TCTA provides a more faithful reconstruction compared with TVecA. This improvement can be attributed to the fact that TCTA exploits the data correlations more effectively.

To further compare the performance, we conduct experiments on both targets using TCTA and TVecA under URS sampling with varying sampling rates. The image quality is evaluated using RMSE and presented in Fig. 16. Apparently, compared with the structured sparsity of EUR or zigzag sampling, the random sparsity in URS sampling is better addressed by TCTA than by TVecA, with the performance gap being more pronounced for the more complex Target 2.

Based on the above ablation experiments, we conclude that TD-TVecA/TCTA is a balanced and validated solution. The truncated-DCT improves computational efficiency, while the Toeplitz matrix methods TVecA/TCTA ensure strong performance and generalization across various sparse sampling strategies. Their combination endows the algorithm with great potential for high-speed and high-quality mmWave 3-D SAR imaging applications.

V. CONCLUSION

This article presents and validates a structured sparse scanning pattern EUR and a corresponding reconstruction algorithm TD-TVecA/TCTA in a practical mmWave 3-D SAR imaging system. The proposed strategy EUR, enables substantial reductions in data acquisition time while maintaining compatibility with mechanical scanning systems. The algorithm TD-TVecA/TCTA, first incorporates the truncated-DCT to compress the structured sparse SAR data induced by EUR. It further reconstructs the compressed

data using the Toeplitz matrix methods TVecA and TCTA, where two novel Toeplitz transformations are applied to the entire matrix. The proposed EUR and TD-TVecA/TCTA are validated on a practical mmWave SAR imaging system that can achieve millimeter-level resolution for penetration imaging. Experimental results demonstrate that the EUR reduces data acquisition time from 17 min to 8–4 min, while the TD-TVecA/TCTA enables high-fidelity reconstruction and imaging within 5 s across various targets and sampling rates.

Nevertheless, the proposed approach has some limitations. First, theoretically, although the suitability of the proposed TCT/TVec constructions for EUR-induced structured sparse pattern is discussed, a rigorous mathematical justification remains to be further explored. Second, experimentally, as the method relies on the low-rank assumption, imaging artifacts of the complex target still require improvement. Third, computationally, although truncated-DCT halves the reconstruction time, the method remains time-consuming when extended to larger scale data (e.g., larger apertures, higher resolutions, and multiple range slices). Fourth, operationally, the experiments are conducted on a static platform, whereas practical scenarios may involve adverse weather conditions, motion-induced trajectory deviations, and irregular measurement noise. These limitations motivate future work on more rigorous theoretical analysis, improved reconstruction robustness for complex scenes, scalable algorithms for large-scale data, and practical deployment in dynamic environments.

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